Research Analysis Report

Research Topic: string

Total Papers: 5

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Executive Summary

This report provides a technical overview of 'string' including methodologies, research gaps and experiment planning.

Literature Review

Introduction: This literature review summarizes findings from 5 research papers related to the selected topic. It highlights research trends, methodologies, key contributions, future directions, and overall observations.

Paper Summaries

• {'title': 'The STRING database in 2023: protein–protein association networks and functional enrichment analyses for any sequenced genome of interest', 'year': 2022, 'summary': 'Abstract Much of the complexity within cells arises from functional and regulatory interactions among proteins. The core of these interactions is increasingly known, but novel interactions continue to be discovered, and the information remains scattered across different database resources, experimental modalities and levels of mechanistic detail. The STRING database (https://string-db.org/) systematically collects and integrates protein–protein interactions—both physical interactions as well as functional associations. The data originate from a number of sources: automated text mining of the scientific literature, computational interaction predictions from co-expression, conserved genomic context, databases of interaction experiments and known complexes/pathways from curated sources. All of these interactions are critically assessed, scored, and subsequently automatically transferred to less well-studied organisms using hierarchical orthology information. The data can be accessed via the website, but also programmatically and via bulk downloads. The most recent developments in STRING (version 12.0) are: (i) it is now possible to create, browse and analyze a full interaction network for any novel genome of interest, by submitting its complement of encoded proteins, (ii) the co-expression channel now uses variational auto-encoders to predict interactions, and it covers two new sources, single-cell RNA-seq and experimental proteomics data and (iii) the confidence in each experimentally derived interaction is'}

• {'title': 'The STRING database in 2021: customizable protein–protein networks, and functional characterization of user-uploaded gene/measurement sets', 'year': 2020, 'summary': "Abstract Cellular life depends on a complex web of functional associations between biomolecules. Among these associations, protein–protein interactions are particularly important due to their versatility, specificity and adaptability. The STRING database aims to integrate all known and predicted associations between proteins, including both physical interactions as well as functional associations. To achieve this, STRING collects and scores evidence from a number of sources: (i) automated text mining of the scientific literature, (ii) databases of interaction experiments and annotated complexes/pathways, (iii) computational interaction predictions from co-expression and from conserved genomic context and (iv) systematic transfers of interaction evidence from one organism to another. STRING aims for wide coverage; the upcoming version 11.5 of the resource will contain more than 14 000 organisms. In this update paper, we describe changes to the text-mining system, a new scoring-mode for physical interactions, as well as extensive user interface features for customizing, extending and sharing protein networks. In addition, we describe how to query STRING with genome-wide, experimental data, including the automated detection of enriched functionalities and potential biases in the user's query data. The STRING resource is available online, at https://string-db.org/."}

- {'title': 'STRING v11: protein–protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets', 'year': 2018, 'summary': 'Abstract Proteins and their functional interactions form the backbone of the cellular machinery. Their connectivity network needs to be considered for the full understanding of biological phenomena, but the available information on protein–protein associations is incomplete and exhibits varying levels of annotation granularity and reliability. The STRING database aims to collect, score and integrate all publicly available sources of protein–protein interaction information, and to complement these with computational predictions. Its goal is to achieve a comprehensive and objective global network, including direct (physical) as well as indirect (functional) interactions. The latest version of STRING (11.0) more than doubles the number of organisms it covers, to 5090. The most important new feature is an option to upload entire, genome-wide datasets as input, allowing users to visualize subsets as interaction networks and to perform gene-set enrichment analysis on the entire input. For the enrichment analysis, STRING implements well-known classification systems such as Gene Ontology and KEGG, but also offers additional, new classification systems based on high-throughput text-mining as well as on a hierarchical clustering of the association network itself. The STRING resource is available online at https://string-db.org/.'}

- {'title': 'The STRING database in 2017: quality-controlled protein–protein association networks, made broadly accessible', 'year': 2016, 'summary': 'A system-wide understanding of cellular function requires knowledge of all functional interactions between the expressed proteins. The STRING database aims to collect and integrate this information, by consolidating known and predicted protein–protein association data for a large number of organisms. The associations in STRING include direct (physical) interactions, as well as indirect (functional) interactions, as long as both are specific and biologically meaningful. Apart from collecting and reassessing available experimental data on protein–protein interactions, and importing known pathways and protein complexes from curated databases, interaction predictions are derived from the following sources: (i) systematic co-expression analysis, (ii) detection of shared selective signals across genomes, (iii) automated text-mining of the scientific literature and (iv) computational transfer of interaction knowledge between organisms based on gene orthology. In the latest version 10.5 of STRING, the biggest changes are concerned with data dissemination: the web frontend has been completely redesigned to reduce dependency on outdated browser technologies, and the database can now also be queried from inside the popular Cytoscape software framework. Further improvements include automated background analysis of user inputs for functional enrichments, and streamlined download options. The STRING resource is available online, at http://string-db.org/.'}

- {'title': 'STRING v10: protein–protein interaction networks, integrated over the tree of life', 'year': 2014, 'summary': 'The many functional partnerships and interactions that occur between proteins are at the core of cellular processing and their systematic characterization helps to provide context in molecular systems biology. However, known and predicted interactions are scattered over multiple resources, and the available data exhibit notable differences in terms of quality and completeness. The STRING database (http://string-db.org) aims to provide a critical assessment and integration of protein–protein interactions, including direct (physical) as well as indirect (functional) associations. The new version 10.0 of STRING covers more than 2000 organisms, which has necessitated novel, scalable algorithms for transferring interaction information between organisms. For this purpose, we have introduced hierarchical and self-consistent orthology annotations for all interacting proteins, grouping the proteins into families at various levels of phylogenetic resolution. Further improvements in version 10.0 include a completely redesigned prediction pipeline for inferring protein–protein associations from co-expression data, an API interface for the R computing environment and improved statistical analysis for enrichment tests in user-provided networks.'}

Research Trends

- abstract
- aims
- among
- arise
- association
- backbone
- between
- biological
- biomolecule
- cells
- cellular
- characterization
- collect
- complex
- complexity
- connectivity
- considered
- context
- continue
- core
- database
- depend
- expressed
- form
- full
- function
- functional
- helps
- important
- increasingly
- interaction
- knowledge
- known
- life
- machinery
- many
- molecular

- much
- needs
- network
- occur
- particularly
- partnership
- processing
- protein
- regulatory
- require
- string
- system-wide
- systematic
- understanding
- versatility
- within

Research Gaps

None

Future Scope

None

Conclusion: Overall, the reviewed studies demonstrate significant progress in this research area while highlighting several opportunities for future work and further experimentation.

Methodology Comparison

Total Papers: 5

Comparison

• {'title': 'The STRING database in 2023: protein–protein association networks and functional enrichment analyses for any sequenced genome of interest', 'research_area': 'General Artificial Intelligence', 'research_problem': 'Abstract Much of the complexity within cells arises from functional and regulatory interactions among proteins', 'methodology': ['System'], 'keywords': ['abstract', 'much', 'complexity', 'within', 'cells', 'arise', 'functional', 'regulatory', 'interaction', 'among', 'protein', 'core', 'increasingly', 'known', 'continue'], 'year': 2022, 'citation_count': 7548, 'quality_score': 90, 'quality_classification': 'Excellent'}

• {'title': 'The STRING database in 2021: customizable protein–protein networks, and functional characterization of user-uploaded gene/measurement sets', 'research_area': 'General Artificial Intelligence', 'research_problem': 'Abstract Cellular life depends on a complex web of functional associations between biomolecules', 'methodology': ['System', 'Rag'], 'keywords': ['abstract', 'cellular', 'life', 'depend', 'complex', 'functional', 'association', 'between', 'biomolecule', 'among', 'protein', 'interaction', 'particularly', 'important', 'versatility'], 'year': 2020, 'citation_count': 6538,

'quality_score': 80, 'quality_classification': 'Very Good'}

• {'title': 'STRING v11: protein–protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets', 'research_area': 'General Artificial Intelligence', 'research_problem': 'Abstract Proteins and their functional interactions form the backbone of the cellular machinery', 'methodology': ['System'], 'keywords': ['abstract', 'protein', 'functional', 'interaction', 'form', 'backbone', 'cellular', 'machinery', 'connectivity', 'network', 'needs', 'considered', 'full', 'understanding', 'biological'], 'year': 2018, 'citation_count': 15211, 'quality_score': 80, 'quality_classification': 'Very Good'}

• {'title': 'The STRING database in 2017: quality-controlled protein–protein association networks, made broadly accessible', 'research_area': 'General Artificial Intelligence', 'research_problem': 'A system-wide understanding of cellular function requires knowledge of all functional interactions between the expressed proteins', 'methodology': ['Framework', 'System'], 'keywords': ['system-wide', 'understanding', 'cellular', 'function', 'require', 'knowledge', 'functional', 'interaction', 'between', 'expressed', 'protein', 'string', 'database', 'aims', 'collect'], 'year': 2016, 'citation_count': 6529, 'quality_score': 80, 'quality_classification': 'Very Good'}

• {'title': 'STRING v10: protein–protein interaction networks, integrated over the tree of life', 'research_area': 'General Artificial Intelligence', 'research_problem': 'The many functional partnerships and interactions that occur between proteins are at the core of cellular processing and their systematic characterization helps to provide context in molecular systems biology', 'methodology': ['System', 'Algorithm'], 'keywords': ['many', 'functional', 'partnership', 'interaction', 'occur', 'between', 'protein', 'core', 'cellular', 'processing', 'systematic', 'characterization', 'helps', 'context', 'molecular'], 'year': 2014, 'citation_count': 9790, 'quality_score': 80, 'quality_classification': 'Very Good'}

Highest Cited Paper

title: STRING v11: protein–protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets

authors: ['Damian Szklarczyk', 'Annika L. Gable', 'D. Lyon', 'Alexander Junge', 'S. Wyder', 'J. Huerta-Cepas', 'M. Simonovic', 'N. Doncheva', 'J. Morris', 'P. Bork', 'L. Jensen', 'C. V. Mering']

year: 2018

citation_count: 15211

summary: Abstract Proteins and their functional interactions form the backbone of the cellular machinery. Their connectivity network needs to be considered for the full understanding of biological phenomena, but the available information on protein–protein associations is incomplete and exhibits varying levels of annotation granularity and reliability. The STRING database aims to collect, score and integrate all publicly available sources of protein–protein interaction information, and to complement these with computational predictions. Its goal is to achieve a comprehensive and objective global network, including direct (physical) as well as indirect (functional) interactions. The latest version of STRING (11.0) more than doubles the number of organisms it covers, to 5090. The most important new feature is an option to upload entire, genome-wide datasets as input, allowing users to visualize subsets as interaction networks and to perform gene-set enrichment analysis on the entire input. For the enrichment analysis, STRING implements well-known classification systems such as Gene Ontology and KEGG, but also offers additional, new classification systems based on high-throughput text-mining as well as on a hierarchical clustering of the association network itself. The STRING resource is available online at https://string-db.org/.

research_problem: Abstract Proteins and their functional interactions form the backbone of the cellular machinery

methodology: ['System']

key_contributions: []

future_work: []

keywords: ['abstract', 'protein', 'functional', 'interaction', 'form', 'backbone', 'cellular', 'machinery', 'connectivity', 'network', 'needs', 'considered', 'full', 'understanding', 'biological']

research_area: General Artificial Intelligence

paper_score: 80

paper_quality: Very Good

Latest Paper

title: The STRING database in 2023: protein–protein association networks and functional enrichment analyses for any sequenced genome of interest

authors: ['Damian Szklarczyk', 'Rebecca Kirsch', 'Mikaela Koutrouli', 'Katerina C. Nastou', 'Farrokh Mehryary', 'Radja Hachilif', 'Annika L. Gable', 'Tao Fang', 'N. Doncheva', 'Sampo Pyysalo', 'P. Bork', 'L. Jensen', 'C. V. Mering']

year: 2022

citation_count: 7548

summary: Abstract Much of the complexity within cells arises from functional and regulatory interactions among proteins. The core of these interactions is increasingly known, but novel interactions continue to be discovered, and the information remains scattered across different database resources, experimental modalities and levels of mechanistic detail. The STRING database (https://string-db.org/) systematically collects and integrates protein–protein interactions—both physical interactions as well as functional associations. The data originate from a number of sources: automated text mining of the scientific literature, computational interaction predictions from co-expression, conserved genomic context, databases of interaction experiments and known complexes/pathways from curated sources. All of these interactions are critically assessed, scored, and subsequently automatically transferred to less well-studied organisms using hierarchical orthology information. The data can be accessed via the website, but also programmatically and via bulk downloads. The most recent developments in STRING (version 12.0) are: (i) it is now possible to create, browse and analyze a full interaction network for any novel genome of interest, by submitting its complement of encoded proteins, (ii) the co-expression channel now uses variational auto-encoders to predict interactions, and it covers two new sources, single-cell RNA-seq and experimental proteomics data and (iii) the confidence in each experimentally derived interaction is

research_problem: Abstract Much of the complexity within cells arises from functional and regulatory interactions among proteins

methodology: ['System']

key_contributions: ['The most recent developments in STRING (version 12']

future_work: []

keywords: ['abstract', 'much', 'complexity', 'within', 'cells', 'arise', 'functional', 'regulatory', 'interaction', 'among', 'protein', 'core', 'increasingly', 'known', 'continue']

research_area: General Artificial Intelligence

paper_score: 90

paper_quality: Excellent

Common Methodologies

• Algorithm

• Framework

• Rag

• System

Research Areas

• General Artificial Intelligence

Common Keywords

• abstract
• aims
• among
• arise
• association
• backbone
• between
• biological
• biomolecule
• cells
• cellular
• characterization
• collect
• complex
• complexity
• connectivity
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• known

- life
- machinery
- many
- molecular
- much
- needs
- network
- occur
- particularly
- partnership
- processing
- protein
- regulatory
- require
- string
- system-wide
- systematic
- understanding
- versatility
- within

Methodology Frequency

System: 5

Rag: 1

Framework: 1

Algorithm: 1

Most Common Methodology: System

Quality Distribution

Excellent: 1

Very Good: 4

Publication Years

2014: 1

2016: 1

2018: 1

2020: 1

2022: 1

Citation Statistics

highest: 15211

lowest: 6529

average: 9123.2

total: 45616

Comparison Summary

total_papers: 5

most_common_methodology: System

research_areas: ['General Artificial Intelligence']

quality_distribution: {'Excellent': 1, 'Very Good': 4}

citation_statistics: {'highest': 15211, 'lowest': 6529, 'average': 9123.2, 'total': 45616}

highest_cited_title: STRING v11: protein–protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets

latest_paper_title: The STRING database in 2023: protein–protein association networks and functional enrichment analyses for any sequenced genome of interest

Research Gap Analysis

Total Papers: 5

Research Areas

• General Artificial Intelligence

Common Keywords

• abstract

• aims

• among

• arise

• association

• backbone

• between

• biological

• biomolecule

• cells

• cellular

• characterization

• collect

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- partnership
- processing
- protein
- regulatory
- require
- string
- system-wide
- systematic
- understanding
- versatility

• within

Future Work

None

Research Area Distribution

General Artificial Intelligence: 5

Keyword Frequency

functional: 5

interaction: 5

protein: 5

cellular: 4

abstract: 3

between: 3

among: 2

core: 2

understanding: 2

much: 1

complexity: 1

within: 1

cells: 1

arise: 1

regulatory: 1

increasingly: 1

known: 1

continue: 1

life: 1

depend: 1

complex: 1

association: 1

biomolecule: 1

particularly: 1

important: 1

versatility: 1

form: 1

backbone: 1

machinery: 1

connectivity: 1

network: 1

needs: 1

considered: 1

full: 1

biological: 1

system-wide: 1

function: 1

require: 1

knowledge: 1

expressed: 1

string: 1

database: 1

aims: 1

collect: 1

many: 1

partnership: 1

occur: 1

processing: 1

systematic: 1

characterization: 1

helps: 1

context: 1

molecular: 1

Research Trends

- functional
- interaction
- protein
- cellular
- abstract
- between
- among
- core
- understanding
- much
- complexity
- within
- cells

• arise
• regulatory

Gap Categories

datasets: []

models: []

evaluation: []

applications: []

general: []

Emerging Topics

• functional
• interaction
• protein
• cellular
• abstract
• between
• among
• core
• understanding

Recommendations

• Investigate unexplored research directions.

Summary

dominant_area: General Artificial Intelligence

top_trend: functional

future_work_items: 0

Experiment Plan

Research Areas

• General Artificial Intelligence

Recommended Datasets

None

Baseline Models

None

Evaluation Metrics

• Accuracy

• Precision

• Recall

• F1-Score

• ROC-AUC

Hardware Requirements

cpu: Intel Core i7 / AMD Ryzen 7

ram: 16 GB

gpu: NVIDIA RTX 3060 or better

storage: 100 GB SSD

Validation Strategy

• Train-Test Split

• K-Fold Cross Validation

• Hyperparameter Tuning

• Statistical Significance Testing

Experimental Workflow

• Collect Dataset

• Preprocess Data

• Train Baseline Models

• Train Proposed Model

• Evaluate Performance

• Compare Results

• Analyze Errors

• Draw Conclusions

Report Summary

Dominant Research Area: General Artificial Intelligence

Top Research Trend: functional

Highest Cited Paper: STRING v11: protein–protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets

Latest Paper: The STRING database in 2023: protein–protein association networks and functional enrichment analyses for any sequenced genome of interest

Recommended Datasets

None

Baseline Models

None

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

Conclusion

The report has been generated using a personalized multi-agent pipeline. The explanation style was adapted according to the Intermediate profile to improve readability and usefulness.